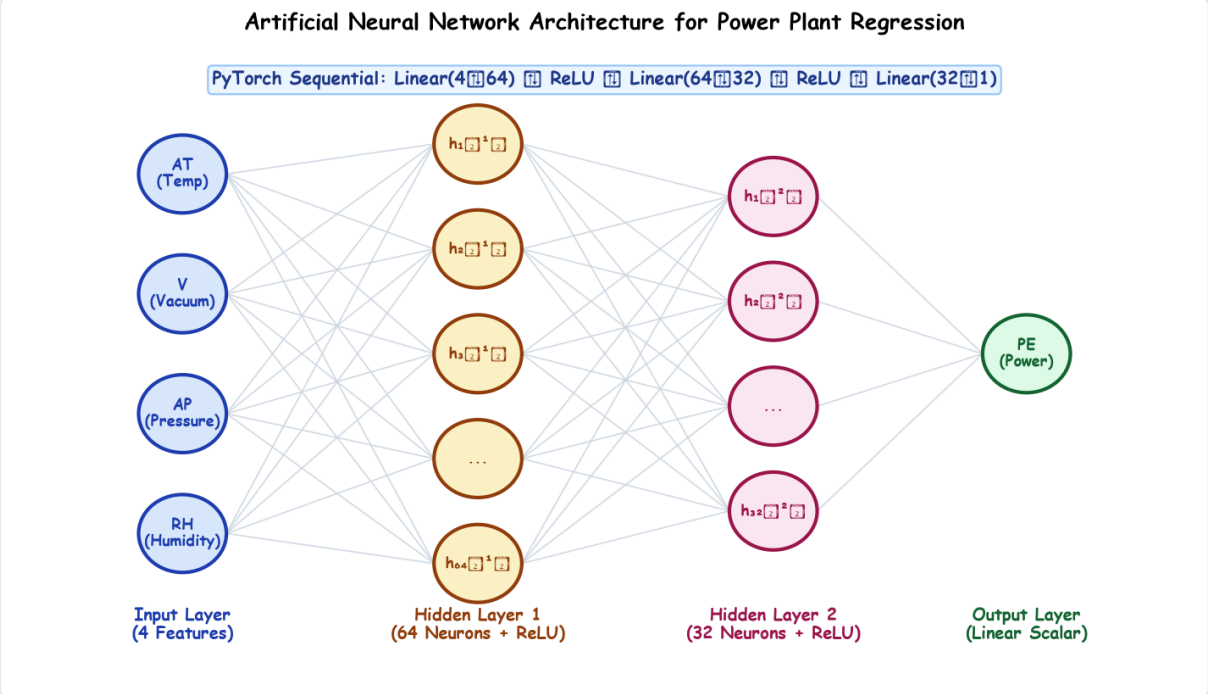
Submodule 38.01: Artificial Neural Networks for Regression (Hinglish)

1. Parichay aur Real-World Problem Formulation

Classical Machine Learning algorithms (jaise Linear Regression ya Support Vector Regression) ke baad, ab hum Deep Learning aur **Artificial Neural Networks (ANN)** ko ek real-world industrial project par implement karenge: **Combined Cycle Power Plant ka Energy Output (PE) Predict Karna**.

Ek Power Plant me gas aur steam turbines chalti hain jo ambient environment (taapmaan, pressure, humidity, vacuum) ke hisab se electricity generate karti hain. Hum 4 inputs se continuous electrical power (PE) predict karenge.



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2. Dataset ke Columns (Features)

Column	Naam	Physical Unit	Description
`AT`	Ambient Temperature	$^\circ C$	Power plant ke aas-paas ka taapmaan.

Column	Naam	Physical Unit	Description
`V`	Exhaust Vacuum	cm Hg	Steam turbine ka vacuum pressure.
`AP`	Ambient Pressure	mbar	Hawa ka atmospheric barometric pressure.
`RH`	Relative Humidity	%	Hawa me nami ka percentage.
`PE` **(Target)**	**Net Electrical Output**	MW	**Total generated power (MegaWatts me) jo predict karni hai.**

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3. Neural Network ka Architecture

1. **Input Layer**: 4 neurons (AT, V, AP, RH).
2. **Hidden Layer 1**: 64 neurons + **ReLU** activation.
3. **Hidden Layer 2**: 32 neurons + **ReLU** activation.
4. **Output Layer**: 1 neuron (**Koi activation function nahi**, bilkul linear). Regression me output par Sigmoid ya ReLU nahi lagate taaki continuous number predict ho sake.

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4. PyTorch Lifecycle ke 6 Steps

1. **Data Load karna**: Pandas se CSV read karna aur null values check karna.
2. **Scaling aur Tensors**: `StandardScaler` lagana aur PyTorch `Float32` tensors me convert karna.
3. **TensorDataset aur DataLoader**: 32-32 samples ke mini-batches banana.
4. `nn.Module` **Model Class**: PyTorch me custom ANN class likhna.
5. **Training Loop**: Forward pass $\rightarrow$ MSE Loss $\rightarrow$ Backpropagation $\rightarrow$ Adam Optimizer step.
6. **Model Save aur Evaluation**: Best model save karna aur R^2 score calculate karna.

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5. Companion Code Files

Is submodule ke sath original Jupyter Notebook aur CSV dataset attach kiye gaye hain:

- ANN_Regression.ipynb

- powerplant_data.csv\n